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Inventor(s)

WYATT; Cameron et al.

INTERLOCK ASSEMBLY, AND MULTI-SLIDE PANEL SYSTEM USING SAME

Abstract

An interlock assembly, comprising a strut having a strut extension, a frame member fixedly attached to the strut. An interlock assembly, additionally wherein the strut and the frame member extend longitudinally from a first end to a second end and are configured to retain an end of a panel member. Further a notch is formed through the strut and the frame member at one of the first end or the second end, wherein the notch is configured to receive a plate member of a connecting frame member, and wherein a first portion of the strut extension is to an interior side of the plate member and a second portion of the strut extension is to an exterior side of the plate member. Additionally, a sliding panel system is described that includes the interlock assembly.

Inventors: WYATT; Cameron (Gilbert, AZ), Sabagh; Samir (Miami, FL), Hardecopf; Sarah J. (Gilbert, AZ), Daes; Daniel (Key Biscayne, FL)

Applicant: Tecnoglass, Inc. (Miami, FL)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Application No. 63/558,542, filed Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure generally relates to a multiple panel (e.g., window or door) system, with at least one sliding panel, and an interlock assembly for use therein.

BACKGROUND

[0003] Many current multi-slide panel systems have an interlock assembly formed using two interlocks, each on a different frame track plane, with their interaction occurring at a 180° offset. One issue with using a system that requires two non-symmetrical interlocks is that more parts are required (a unique interior or unique exterior interlock), or additional assembly steps are required to transform them from interior or exterior interlocks. Further, to make an interlock symmetric results in a non-optimal strut placement location. This non-symmetrical system currently in place can result in extended amounts of labor, cost of materials and can result in a lack of thermal efficiency or sealing efficiency. This makes the current non-symmetrical interlock systems currently in place a bad choice for modern multi-slide panel systems.

SUMMARY

[0004] The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

[0005] In some aspects, the techniques described herein relate to an interlock assembly including a strut having a strut extension and a frame member fixedly attached to the strut. The strut and the frame member extend longitudinally from a first end to a second end and are configured to retain an end of a panel member. The interlock assembly additionally includes a notch formed through the strut and the frame member at one of the first end or the second end, wherein the notch is configured to receive a plate member of a connecting frame member, and wherein a first portion of the strut extension is to an interior side of the plate member and a second portion of the strut extension is to an exterior side of the plate member.

[0006] In some aspects, the present disclosure includes a sliding panel system having a first panel member positioned on a first track of the sliding panel system and a second panel member positioned on a second track of the sliding panel system, wherein the second panel member is movable on the second track relative to the first panel member. The system further includes a first interlock assembly on the first panel member including a first strut having a first strut extension and a first frame member fixedly attached to the first strut. The first strut and the first frame member extend longitudinally from a first end to a second end and are configured to retain an end of a first panel member. The first interlock assembly further includes a first notch formed through the first strut and the first frame member at one of the first end or the second end, wherein the first notch is configured to receive a first plate member of a first connecting frame member, and wherein a first portion of the first strut extension is to an interior side of the first plate member and a second portion of the first strut extension is to an exterior side of the first plate member. The system additionally includes a second interlock assembly on the second panel member including a second

strut having a second strut extension and a second frame member fixedly attached to the second strut. The second strut and the second frame member extend longitudinally from a first end to a second end and are configured to retain an end of a second panel member. and the second interlock assembly further includes a second notch formed through the second strut and the second frame member at one of the first end or the second end, wherein the second notch is configured to receive a second plate member of a second connecting frame member. Also, a first portion of the second strut extension is to an interior side of the second plate member and a second portion of the second strut extension is to an exterior side of the second plate member. Further, respective strut hook members of the first interlock assembly and the second interlock assembly are configured to seal a gap between the first panel member and the second panel member when the second panel member is moved to a closed position.

[0007] These and additional aspects of the present disclosure are described in more detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a front view of an example of a multi-slide panel system including a sill assembly according to the present aspects;

[0009] FIG. 2 is a right side view of the multi-slide panel system of FIG. 1;

[0010] FIG. 3 is a partial top, cross-sectional view of the multi-slide panel system of FIG. 1, with the longitudinal length of each panel member abbreviated and with the sill assembly removed for the sake of clarity, wherein a gap between adjacent panel members is sealed with a corresponding pair of symmetric interlock assemblies;

[0011] FIG. 4 is a top cross-sectional view of one of the interlock assemblies of FIGS. 1-3;

[0012] FIG. 5 is an enlarged, magnified partial top, cross-sectional view the dashed line area in FIG. 3, including of the engagement of opposing interlock assemblies;

[0013] FIG. 6 is an isometric view of a portion of a panel member including a cut-away portion of an interlock assembly of FIGS. 1 and 3-5;

[0014] FIG. 7 is an exploded perspective view of a panel member of the multi-slide panel system of FIG. 1 and corresponding frame members, including the interlock assembly as described herein; and

[0015] FIG. 8 is an enlarged, front left perspective view of the dashed line area in FIG. 7, including a portion of the interlock assembly described herein having a notched area for connection with a top frame member of the multi-slide panel system of FIG. 1.

DETAILED DESCRIPTION

[0016] Various aspects of the disclosure are now described with reference to the drawings, wherein like reference numerals are used to refer to elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to promote a thorough understanding of one or more aspects of the disclosure. It may be evident in some or all instances, however, that any aspects described below can be practiced without adopting the specific design details described below.

[0017] An interlock assembly is an item of a sliding or multi-sliding door or window system that provides a seal between two different panels on different tracks, meaning a seal to another panel that is one track width distance away. Interlock assemblies are designed to work in pairs, meaning an interlock system will include two interlock assemblies, each on a different frame track plane, with their interaction occurring at a 180° offset.

[0018] Aspects of the disclosure include an interlock assembly for a multi-panel system, wherein the interlock assembly is configured as a symmetric interlock assembly such that each interlock assembly may be placed on any side of a panel to accommodate various different panel

configurations. Each interlock assembly will be attachable to a panel on at least one side of the panel and will movably mate with a corresponding interlock assembly on an opposing side of adjacent panel when the panels move into a fully closed position. Each interlock assembly includes a strut and a frame member which together work to attach the interlock assembly to a panel member.

[0019] The frame member includes a first crimp member on the interlocking side and a second crimp member on the side opposite the interlocking side. The first crimp member and second crimp member each hold an extrusion of the strut to hold the strut in place in the frame member. The strut and frame member each further include a strut hook member and a frame member wall member respectively. The strut first wall member extends along the portion of the panel member adjacent to an opposing panel and the frame member first wall member extends along the opposing side of the panel member to create a U-shape around the panel member and to thereby affix the interlock assembly to the panel member. Additionally the strut and frame member may be additionally affixed to the panel member through a securing means into the side of the panel.

[0020] Thus, the disclosed interlock assembly provides a multi-slide panel system with an interlock assembly that is symmetrical and easier to produce and install, and requiring reduced inventory, when compared to existing solutions.

[0021] Referring to FIGS. 1-3, according to one example, a multi-slide panel system **101** includes a first panel **103**, a second panel **105**, and a third panel **107** each including at least one interlock assembly **100** of an opposing pair of interlock assemblies **125** to provide a vertically-extending seal between two adjacent panels, at least one of which is movable relative to the other. The panels **103**, **105**, and **107** are supported within a frame system that includes a multi-track sill assembly **122** at a bottom side, a multi-track top frame **109** at a top side, and opposing end frames **111** and **113** at each end. The panels **103**, **105**, and **107** of the multi-slide panel system **101** may be, for example, doors and/or windows. At least one, and preferably two, of the first panel **103**, the second panel **105**, and the third panel **107** is/are slidably movable with the frame system. For instance, the first panel **103** may be a fixed panel, and the second panel **105** may be slidable (e.g., to the left in the example of FIG. 1) to overlap with the first panel **103**, and the third panel **107** may be slidable (e.g., to the left in the example of FIG. 1) to overlap with both the first panel **103** and the second panel **105**. The multi-slide panel system **101** may be configured with relatively narrow frame members, compared to prior solutions, to provide a sleek, modern look that minimizes the reveal widths, e.g., a fixed panel reveal **115**, an interlock reveal **117**, a lead panel reveal **119**, a top rail reveal **121**, and a bottom rail reveal **123**, thereby maximizing a view through the panels **103**, **105** and **107**. The panels **103**, **105** and **107** may include a single or multi-pane sheet of a transparent, translucent, or non-transparent material, such as but not limited to a glass, a plastic, or a wood material. In a typical example, the panels **103**, **105** and **107** include two or three panes of glass, which can be sealed and filled with an insulating material therebetween, such as a gas having a lower thermal conductivity than air. An example of one such gas includes argon. In any case, the interlock assembly **100** of the multi-slide panel system **101** provides an improvement over current solutions, as the interlock assembly **100** of the present disclosure is symmetric. In other words, the same interlock assembly **100** can be used on each of the opposing pair of interlocker assemblies **125**. For example, referring to FIG. 3, the interlock assembly **100** on the back, right side of panel member **105** is identical to the interlock assembly **100** on the front left side of panel member **107**, e.g., one interlock assembly **100** is rotated 180 degrees about its longitudinal axis relative to the other one. Having a symmetric interlock assembly **100** reduces the labor required to manufacture the interlock assembly, and reduces the required amount of inventory (since all interlock assemblies are the same). One or more other aspects of the interlock assembly **100** also provide improvements over existing solutions, such as but not limited to: a mounting extension on a strut of the interlock assembly having a large area that aids in certain aspects with connecting to other frame members; a hybrid configuration of a metal frame member and non-metal strut, which reduces thermal energy

transfer; strategically located crimp locations to avoid crimp lines from being within sight lines; and, a relatively narrow interlock reveal, as compared to existing solutions. Each of these features is discussed in more detail below.

[0022] Referring to FIGS. 4-7, in the example implementation of FIGS. 1-3, which should not be construed as limiting, the example interlock assembly **100** includes a strut **102** fixedly positioned within a frame member **104**. The frame member **104** includes a first crimp member **106** and a second crimp member **108**, which hold or mate with a first strut extension **110** and a second strut extension **112** respectively to hold the strut within the frame member **104**. The strut **102** further includes a strut base member **114** that extends across a thickness or depth **127** of the respective panel member, and a strut hook member **116** having a hook shape that supports a face of a respective panel member on one side and that forms an interlocking groove for engagement and sealing with an opposing interlock assembly. The frame member **104** further includes a frame base member **118** that extends along the thickness or depth **127** of the respective panel member, and a frame wall member **120** that opposes the strut hook member **116** to cooperatively support the respective panel member. The strut base member **114** and strut hook member **116** form a first L-shape, and the frame base member **118** and frame wall member **120** form an opposite L-shape. The strut base member **114** and frame base member **118** each extend adjacent to one another in a parallel direction. The strut hook member **116** extends in a perpendicular direction, relative to the strut base member **114** and frame base member **118**, on one side of the interlock assembly **100**, and the frame wall member **120** extends in a perpendicular direction on an opposing side to define a panel-retaining channel **129**. As such, the interlock assembly **100** has a U-shaped (or inverted U-shaped) cross-section when the strut **102** and frame member **104** are attached to one another.

[0023] The panel-retaining channel **129** is configured to fixedly hold a respective one of the panel members **103**, **105**, or **107** such that the strut hook member **116** will extend along the edge of a first face of the respective panel member and the frame wall member **120** will extend along the edge of a second face of the respective panel member. Advantageously, as the strut **102** is configured to have a lower thermal conductivity than the frame member **104**, the configuration of the strut **102** covering the end and a part of an interior side of a respective panel cuts off a thermal path through the frame member **104**, thereby providing the interlock assembly **100** with improved thermal performance.

[0024] Referring specifically to FIG. 7, as an interlock assembly **100** extends the full, vertical length at least end of a respective panel member, such as the left end of panel member **107** in this example, the interlock assembly **100** thereby creates a U-shape around a full length vertical edge of the respective panel member **107**. The interlock assembly **100** can be further affixed to retain the panel member **107** by inserting a securing mechanism **131**, such as a screw or bolt, through the frame base member **118** and further through the strut base member **114** into a top rail sub-assembly **133** and a bottom rail/roller sub-assembly **135** of the panel member **107**. Additionally, in this case, the framing of the panel member **107** is completed by an end frame member **137** affixed to the top rail sub-assembly **133** and the bottom rail/roller sub-assembly **135** by the securing mechanism **131**. Additionally, in an aspect, the securing mechanism **131** may be inserted along line **134-134** (see FIG. 4, and also see FIG. 7) such that the securing mechanism **131** is secured through a full thickness of the strut base member **114** in order to provide a secure connection.

[0025] Referring back to FIG. 4, as discussed above, the strut **102** is held within the frame member **104** through the use of the first crimp member **106** and the second crimp member **108** of the frame member **104**, each of which includes a substantially non-elastically deformable tab that is crimped to hold or mate with the first strut extension **110** and the second strut extension **112**, respectively. The first crimp member **106** of the frame member **104** is located at the distal end of the frame member base member **118** and the second crimp member **108** is located in the frame wall member **120** at a point adjacent to a location where the frame base member **118** and the frame wall member **120** meet. The first strut extension of **110** is located at the distal end of the strut base member **114**

and mates with the second crimp member **108**, and the second strut extension **112** is located at the opposing end of the strut base member **114** on the outside at the point where the strut base member **114** meets the strut hook member **116**. The second strut extension **112** mates with the second crimp member **108**.

[0026] This described configuration of the strut **102** and the frame member **104** provides numerous benefits, First, the first crimp member **106**, on the interlocking side, or the non-visible side, and the second crimp member **108** located within the frame member **104** may complete the hybrid connection between the strut **102** and frame member **104**. As used herein, the term hybrid connection refers to the difference in materials used to form the interlock assembly **100**, which may provide differences in thermal conductivity. For instance, in an aspect, in order to reduce the transfer of thermal energy from one side (such as an exterior side) to another side (such as an interior side) of the interconnect assembly **100**, the strut **102** have a first thermal conductivity that is less than a second thermal conductivity of frame member **104**. For example, the strut **102** may be made from a plastic material, such as but not limited to a polyamide, while the frame member **104** made be made from a metal material, such as but not limited to an aluminum. It should be understood, however, that any of a plurality of different materials may be used for each of these components, and in some aspects they both may be made from a same material. Second, a backup roller of a crimping machine can utilize the relatively long frame wall member **120** to crimp the first crimp member **106** without deforming the structure of the frame member **104**. And, another backup roller of the crimping machine utilize the relatively long frame base member **118** to crimp the second crimp member **108** without deforming the structure of the frame member **104**. In some cases, both the first and second crimping operations on the first and second crimp members **106**, **108** can be performed at the same time. This structural stability is advantageous to efficient crimping, as many interlockers bend during the crimping process and require complex strutting steps like supporting the parts or doing multiple passes. Third, the structure described herein allows for the longest possible strut **102**, measured from a distal end of the strut hook member **116** to an opposite distal end of the strut base member **114**, which creates high energy efficiency. Fourth, besides being symmetrical, the interlock assembly **100** of the present disclosure is configured to be a bypass interlock. For example, the interlock assembly **100** has a low profile (e.g., extends a short distance away from the panel) that enables the panel locks or other hardware to bypass or avoid interfering with the interlock assembly **100** when the panels are moved from one position to another. Thus, the interlock assembly **100** described herein provides good manufacturability due to the good crimping support and no bending, excellent thermal properties, and enhanced aesthetics as both crimps are hidden from view.

[0027] In an aspect, the strut hook member **116** includes an interlock member **124** spaced apart from a panel-retaining wall **132** to define an interlock groove **126** configured to sealingly engage with a corresponding interlock groove of an opposing interlock assembly. The interlock member **124** extends from the strut base member **114** to define an outside wall of the strut hook member **116**. Relative to panel-retaining wall **132**, the interlock member **124** may extend at an angle in a range of 10 degrees to 50 degrees, or more preferably in a range of 20 degrees to 40 degrees. This angle and length of the interlock member **124** at least partially defines a depth of the interlock groove **126** located between the interlock member **124** and the strut hook member **116**. This interlock groove **126** is configured to receive the interlock member **124** of an opposing interlock assembly **100** as can be seen in FIG. 5.

[0028] Referring specifically to FIG. 5, for example, when two panel members of a multi-slide panel system **101**, such as panel members **107** and **105** in this example, are being moved to the closed position, a first interlock assembly **100** of panel member **107** will move towards a second interlock assembly **100** of panel member **105**, and the respective interlock members **124** will move into the respective interlock grooves **126** to close the gap between the panel members **107** and **105**. In other words, the respective interlocks **124** will then engage with one another and/or with the

opposing interlock grooves **126**, creating a seal between the respective struts **102** of the respective interlock assemblies **100**.

[0029] Moreover, as mentioned above, the interaction of the opposing interlocking assemblies **100** described above in relation to FIG. **5** provides the maximum combined strut length between the strut **102** of the interlock assembly **100** of panel member **107** and the strut **102** of the interlock assembly **100** of panel member **105**. This extended strut length provides a dramatic increase in the thermal properties of the interlock system. As described previously, in an aspect, the frame member **104** is made of a material such as aluminum or a similar material with good strength, malleability and weather resistance, and it can be made to look good aesthetically, however, aluminum does not include good thermal properties. The strut **102**, on the other hand, located within the frame member **104**, is comprised of a material with good thermal properties such as a rubber or plastic material. It is therefore advantageous to increase the length of the strut as much as possible to increase the thermal efficiency of the interlock assembly **100** and/or the combined interlock assemblies **100**. Additionally the surface of the interlock member **124**, which engages the opposing interlock is strutted, which provides the largest surface area possible for the interlock member **124**, and the connection between one panel to the next has the lowest thermal conductivity possible. Other prior solutions, which do not perform as well as the present system, include designs have metal tangs, and these can have exposure to conduction paths along the interior glass thereby reducing the thermal properties.

[0030] The side of the strut hook member **116** facing away from the respective panel member further includes a groove **128** configured to removably hold a weatherstrip material, which can extend so fill the interlock groove **126**. The weatherstrip groove **128** is located at the distal end of the strut hook member **116**, such as toward a front of, or in front of, the interlock groove **126**. The weather strip groove **128** allows for a weatherstrip to be inserted into the groove, and the weatherstrip can interact with the interlock member **124** of the opposing interlock assembly. This weatherstrip located in the weatherstrip groove **128** can provide additional protection for the interlock assembly **100** from external elements as well as provide additional thermal properties. Finally, the weatherstrip can hide the internal elements of the interlock assembly **100**, thereby increasing the aesthetic appeal of the interlock system.

[0031] Referring specifically to FIG. **7**, the interlock assembly **100** further includes securing openings **130**. The securing openings **130**, may include screw holes for use at the bottom and top of the interlock assembly **100** to attach various components to the interlock assembly **100**. The securing openings **130**, apart from providing additional securing mechanisms to attach additional components to the interlock assembly **100**, the location of these securing openings further improves weatherization, and these securing openings **130** are not visible from the outside and therefore do not reduce the aesthetic appeal.

[0032] Referring to FIGS. **7** and **8**, in an aspect, the interlock assembly **100** is further configured to have a notch **136**, such as an area of the strut **102** and the frame member **104**, removed to accommodate a plate **138** (see FIG. **7**) of the top rail sub-assembly **133** when the interlock assembly **100** is secured to the top rail sub-assembly **133** via the securing mechanism **131**. Advantageously, the first strut extension **110** if the interlock assembly **100** has a depth **140** configured to provide sufficient structural support to both the reduced area **142** of the interlock groove **126**, e.g., to the outside of the plate **138**, as well as the reduced area **144** of the remaining portion of the first strut extension interior to the plate **138**. In other words, the first strut extension **110** overlaps both the interior and exterior of the panel member **107**. As such, due to the size of the first strut extension **110**, even in view of the formation of the notch **136**, the configuration as described herein avoids forming a void (e.g., an opening to the interior) in the notched area, thereby providing an improved thermal design.

[0033] Thus, in an implementation, the present disclosure provides for a high performance symmetric aluminum/polyamide interlock assembly that has unique thermal efficiency, aesthetic,

and manufacturing processing advantages compared to prior solutions.

[0034] In some aspect, when designing an interlock assembly, one or more of the following criteria may be utilized.

[0035] Narrowness in site line and frame depth—Narrowness being defined in dimension corresponding to the interlock reveal **117**. Frame depth being defined as dimension from interior to an exterior of the panel. Customers want both of these minimized. U-factor is improved if narrowness in site line is minimized. Cost can be reduced with a smaller part as well (less material mass).

[0036] Bending strength—as being defined as the force required to bend the interlocker along the plane orthogonal to FIG. 4. More bending strength is optimal, as the door performs better in structural ratings.

[0037] Thermal efficiency/sealing efficiency—as defined by thermal conductivity, air infiltration, and overall system u-factor. This all should be as low as possible.

[0038] Aesthetics and other customer drivers—like finish options, tactile feel, and no visual struts, crimp lines or fastener heads. A smooth anodized aluminum surface would be the baseline for best, with everything else getting worse with any variance from this.

[0039] Symmetry (to reduce inventory)—not having interior exterior interlocks, handed interlocks, or additional covers, screws or features required to move an interlock from interior to exterior except a rotation.

[0040] Manufacturing and machinability concerns—can the profile be ran through standard rolling, crimping, cutting and machining equipment. When notching for the interlock does it leave a void, etc.

[0041] These criteria are sometimes diametrically opposed to each other (example bending strengths with frame depth). More frame depth of material increases bending strength all other things being equal.

[0042] The interlocker assembly of the present disclosure balances these characteristics by using one or more of the features described above.

[0043] Additionally, aspects of the disclosure may include one or more of the following clauses.

[0044] Clause 1. An interlock assembly, comprising: a strut having a strut extension; a frame member fixedly attached to the strut; wherein the strut and the frame member extend longitudinally from a first end to a second end and are configured to retain an end of a panel member; and a notch formed through the strut and the frame member at one of the first end or the second end, wherein the notch is configured to receive a plate member of a connecting frame member, and wherein a first portion of the strut extension is to an interior side of the plate member and a second portion of the strut extension is to an exterior side of the plate member.

[0045] Clause 2. The interlock assembly of clause 1, wherein the strut further comprises a strut hook member having a hook shape configured to sealingly connect with an opposing strut hook member of an opposing interlock assembly, wherein the strut hook member and the opposing strut hook member are identical shapes rotated 180 degree about a longitudinal axis.

[0046] Clause 3. The interlock assembly of any preceding clause, wherein the strut comprises a first material having a first thermal conductivity, and the frame member comprises a second material having a second thermal conductivity, wherein the first thermal conductivity is less than the second thermal conductivity.

[0047] Clause 4. The interlock assembly of any preceding clause, wherein the strut further comprises a panel-retaining wall, and wherein the frame member comprises a frame wall member spaced apart from the panel-retaining wall and defining a panel-retaining channel configured to fixedly hold the end of the panel member.

[0048] Clause 5. The interlock assembly of any preceding clause, wherein the interlock assembly has a symmetrical configuration that can be used on either side of an opposing pair of interlock assemblies.

[0049] Clause 6. The interlock assembly of any preceding clause, wherein a first one of the opposing pair of interlock assemblies is rotated 180 degrees about its longitudinal axis relative to a second one of the opposing pair of interlock assemblies.

[0050] Clause 7. The interlock assembly of any preceding clause, wherein the strut contacts the end and a portion of an interior surface of the panel member and prevents direct contact between the frame member and the end and the portion of an interior surface of the panel member.

[0051] Clause 8. The interlock assembly of any preceding clause, wherein one side of the strut hook member further includes a groove configured to removably hold a weatherstrip material.

[0052] Clause 9. The interlock assembly of any preceding clause, further comprising at least one securing opening, wherein the at least one securing opening includes a screw hole to attach additional components to the interlock assembly.

[0053] Clause 10. A sliding panel system, comprising: a first panel member positioned on a first track of the sliding panel system; a second panel member positioned on a second track of the sliding panel system, wherein the second panel member is movable on the second track relative to the first panel member; a first interlock assembly on the first panel member comprising: a first strut having a first strut extension; a first frame member fixedly attached to the first strut; wherein the first strut and the first frame member extend longitudinally from a first end to a second end and are configured to retain an end of a first panel member; and a first notch formed through the first strut and the first frame member at one of the first end or the second end, wherein the first notch is configured to receive a first plate member of a first connecting frame member, and wherein a first portion of the first strut extension is to an interior side of the first plate member and a second portion of the first strut extension is to an exterior side of the first plate member; and a second interlock assembly on the second panel member comprising: a second strut having a second strut extension; a second frame member fixedly attached to the second strut; wherein the second strut and the second frame member extend longitudinally from a first end to a second end and are configured to retain an end of a second panel member; and a second notch formed through the second strut and the second frame member at one of the first end or the second end, wherein the second notch is configured to receive a second plate member of a second connecting frame member, and wherein a first portion of the second strut extension is to an interior side of the second plate member and a second portion of the second strut extension is to an exterior side of the second plate member; and wherein respective strut hook members of the first interlock assembly and the second interlock assembly are configured to seal a gap between the first panel member and the second panel member when the second panel member is moved to a closed position.

[0054] Clause 11. The sliding panel system of clause 10, wherein the first strut of the first interlock assembly further comprises a first strut hook member having a hook shape configured to sealingly connect with the second strut hook member of the second interlock assembly, wherein the first strut hook member and the second strut hook member are identical shapes rotated 180 degree about a longitudinal axis.

[0055] Clause 12. The sliding panel system of any preceding clause, wherein each strut comprises a first material having a first thermal conductivity, and each frame member comprises a second material having a second thermal conductivity, wherein the first thermal conductivity is less than the second thermal conductivity.

[0056] Clause 13. The sliding panel system of any preceding clause, wherein each strut further comprises a respective panel-retaining wall, and wherein each frame member comprises a respective frame wall member spaced apart from the respective panel-retaining wall and defining a respective panel-retaining channel configured to fixedly hold the end of each panel member.

[0057] Clause 14. The sliding panel system of any preceding clause, wherein each interlock assembly has a symmetrical configuration that can be used on either side of an opposing pair of interlock assemblies.

[0058] Clause 15. The sliding panel system of any preceding clause, wherein a first one of the

opposing pair of interlock assemblies is rotated 180 degrees about its longitudinal axis relative to a second one of the opposing pair of interlock assemblies.

[0059] Clause 16. The sliding panel system of any preceding clause, wherein each strut contacts the end and a portion of an interior surface of the respective panel member and prevents direct contact between the respective frame member and the end and the portion of an interior surface of the respective panel member.

[0060] Clause 17. The sliding panel system of any preceding clause, wherein one side of each strut hook member further includes a groove configured to removably hold a weatherstrip material.

[0061] Clause 18. The sliding panel system of any preceding clause, wherein each interlock assembly further includes at least one securing opening, wherein the at least one securing opening includes a screw hole to attach additional components to the interlock assembly.

[0062] In general, the description of the aspects disclosed should be considered as being illustrative in all respects and not being restrictive. The scope of the present disclosure is shown by the claims rather than by the above description, and is intended to include meanings equivalent to the claims and all changes in the scope. While preferred aspects of the disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure.

Claims

1. An interlock assembly, comprising: a strut having a strut extension; a frame member fixedly attached to the strut; wherein the strut and the frame member extend longitudinally from a first end to a second end and are configured to retain an end of a panel member; and a notch formed through the strut and the frame member at one of the first end or the second end, wherein the notch is configured to receive a plate member of a connecting frame member, and wherein a first portion of the strut extension is to an interior side of the plate member and a second portion of the strut extension is to an exterior side of the plate member.
2. The interlock assembly of claim 1, wherein the strut further comprises a strut hook member having a hook shape configured to sealingly connect with an opposing strut hook member of an opposing interlock assembly, wherein the strut hook member and the opposing strut hook member are identical shapes rotated 180 degree about a longitudinal axis.
3. The interlock assembly of claim 1, wherein the strut comprises a first material having a first thermal conductivity, and the frame member comprises a second material having a second thermal conductivity, wherein the first thermal conductivity is less than the second thermal conductivity.
4. The interlock assembly of claim 1, wherein the strut further comprises a panel-retaining wall, and wherein the frame member comprises a frame wall member spaced apart from the panel-retaining wall and defining a panel-retaining channel configured to fixedly hold the end of the panel member.
5. The interlock assembly of claim 1, wherein the interlock assembly has a symmetrical configuration that can be used on either side of an opposing pair of interlock assemblies.
6. The interlock assembly of claim 5, wherein a first one of the opposing pair of interlock assemblies is rotated 180 degrees about its longitudinal axis relative to a second one of the opposing pair of interlock assemblies.
7. The interlock assembly of claim 1, wherein the strut contacts the end and a portion of an interior surface of the panel member and prevents direct contact between the frame member and the end and the portion of an interior surface of the panel member.
8. The interlock assembly of claim 2, wherein one side of the strut hook member further includes a groove configured to removably hold a weatherstrip material.
9. The interlock assembly of claim 1, further comprising at least one securing opening, wherein the at least one securing opening includes a screw hole to attach additional components to the interlock

assembly.

10. A sliding panel system, comprising: a first panel member positioned on a first track of the sliding panel system; a second panel member positioned on a second track of the sliding panel system, wherein the second panel member is movable on the second track relative to the first panel member; a first interlock assembly on the first panel member comprising: a first strut having a first strut extension; a first frame member fixedly attached to the first strut; wherein the first strut and the first frame member extend longitudinally from a first end to a second end and are configured to retain an end of a first panel member; and a first notch formed through the first strut and the first frame member at one of the first end or the second end, wherein the first notch is configured to receive a first plate member of a first connecting frame member, and wherein a first portion of the first strut extension is to an interior side of the first plate member and a second portion of the first strut extension is to an exterior side of the first plate member; and a second interlock assembly on the second panel member comprising: a second strut having a second strut extension; a second frame member fixedly attached to the second strut; wherein the second strut and the second frame member extend longitudinally from a first end to a second end and are configured to retain an end of a second panel member; and a second notch formed through the second strut and the second frame member at one of the first end or the second end, wherein the second notch is configured to receive a second plate member of a second connecting frame member, and wherein a first portion of the second strut extension is to an interior side of the second plate member and a second portion of the second strut extension is to an exterior side of the second plate member; and wherein respective strut hook members of the first interlock assembly and the second interlock assembly are configured to seal a gap between the first panel member and the second panel member when the second panel member is moved to a closed position.

11. The sliding panel system of claim 10, wherein the first strut of the first interlock assembly further comprises a first strut hook member having a hook shape configured to sealingly connect with the second strut hook member of the second interlock assembly, wherein the first strut hook member and the second strut hook member are identical shapes rotated 180 degree about a longitudinal axis.

12. The sliding panel system of claim 11, wherein each strut comprises a first material having a first thermal conductivity, and each frame member comprises a second material having a second thermal conductivity, wherein the first thermal conductivity is less than the second thermal conductivity.

13. The sliding panel system of claim 10, wherein each strut further comprises a respective panel-retaining wall, and wherein each frame member comprises a respective frame wall member spaced apart from the respective panel-retaining wall and defining a respective panel-retaining channel configured to fixedly hold the end of each panel member.

14. The sliding panel system of claim 10, wherein each interlock assembly has a symmetrical configuration that can be used on either side of an opposing pair of interlock assemblies.

15. The sliding panel system of claim 14, wherein a first one of the opposing pair of interlock assemblies is rotated 180 degrees about its longitudinal axis relative to a second one of the opposing pair of interlock assemblies.

16. The sliding panel system of claim 10, wherein each strut contacts the end and a portion of an interior surface of the respective panel member and prevents direct contact between the respective frame member and the end and the portion of an interior surface of the respective panel member.

17. The sliding panel system of claim 10, wherein one side of each strut hook member further includes a groove configured to removably hold a weatherstrip material.

18. The sliding panel system of claim 10, wherein each interlock assembly further includes at least one securing opening, wherein the at least one securing opening includes a screw hole to attach additional components to the interlock assembly.
